THE COMPOSITION OF STRICT MORPHISMS OF FILTERED OBJECTS IS STRICT UNDER AN INJECTIVITY OR SURJECTIVITY HYPOTHESIS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all three assertions, including the counterexample; each displayed computation is followed by the justification of its individual steps, and the closing sentence is a conclusion rather than a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Strict morphisms of filtered objects. Same chapter, the definition of a strict morphism of filtered objects. Supplies the condition on the induced filtration of the image of a morphism that all three assertions are about.
- Induced and quotient filtrations are strict. Same chapter, Lemma lemma-induced-and-quotient-strict. Supplies that the inclusion and the quotient used in the counterexample are strict morphisms of filtered objects, so that the counterexample really lies inside the hypotheses of part (1).

Source reference. Original lemma 0124: sources/stacks-homology/pinned-original-source.tex, source lines 5020–5020. Navigation pointer invoked at proof line 5050; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $f : A \to B$, $g : B \to C$ be strict morphisms of filtered objects.*

(1) *In general the composition $g \circ f$ is not strict.*

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(2) *If g is injective, then $g \circ f$ is strict.*
(3) *If f is surjective, then $g \circ f$ is strict.*

Proof. Let B a vector space over a field k with basis e_1, e_2, with the filtration $F^nB = B$ for $n < 0$, with $F^0B = ke_1$, and $F^nB = 0$ for $n > 0$. Now take $A = k(e_1 + e_2)$ and $C = B/ke_2$ with filtrations induced by B, i.e., such that $A \to B$ and $B \to C$ are strict (Lemma 0124). Then $F^n(A) = A$ for $n < 0$ and $F^n(A) = 0$ for $n \geq 0$. Also $F^n(C) = C$ for $n \leq 0$ and $F^n(C) = 0$ for $n > 0$. So the (nonzero) composition $A \to C$ is not strict.

Assume g is injective. Then

$$\begin{aligned} g(f(F^pA)) &= g(f(A) \cap F^pB) \\ &= g(f(A)) \cap g(F^p(B)) \\ &= (g \circ f)(A) \cap (g(B) \cap F^pC) \\ &= (g \circ f)(A) \cap F^pC. \end{aligned}$$

The first equality as f is strict, the second because g is injective, the third because g is strict, and the fourth because $(g \circ f)(A) \subset g(B)$.

Assume f is surjective. Then

$$\begin{aligned} (g \circ f)^{-1}(F^iC) &= f^{-1}(F^iB + \mathrm{Ker}(g)) \\ &= f^{-1}(F^iB) + f^{-1}(\mathrm{Ker}(g)) \\ &= F^iA + \mathrm{Ker}(f) + \mathrm{Ker}(g \circ f) \\ &= F^iA + \mathrm{Ker}(g \circ f) \end{aligned}$$

The first equality because g is strict, the second because f is surjective, the third because f is strict, and the last because $\mathrm{Ker}(f) \subset \mathrm{Ker}(g \circ f)$. □